

Project Report

on

Healthcare Content Generator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

Healthcare professionals and patients frequently require structured medical content such as patient summaries, clinical notes, and discharge instructions. Manual creation of such content is time-consuming, inconsistent, and prone to human error.

While Generative AI can automate this process, standalone models often generate hallucinated or contextually incorrect outputs due to lack of domain grounding.

Therefore, a system is needed that:

- Generates structured healthcare content
- Uses domain-specific context for accuracy
- Ensures consistency, clarity, and safety

1.2 Project Objectives

- Develop a Healthcare Content Generator using Generative AI
- Implement Retrieval-Augmented Generation (RAG) for contextual accuracy
- Generate structured outputs such as:
 - Patient summaries
 - Clinical notes
 - Discharge instructions
- Build a user-friendly web interface
- Ensure secure API key handling
- Design a modular and scalable system architecture

1.3 Scope of the Project

In Scope

- Healthcare content generation using AI
- Context retrieval using vector database
- Customizable output formats and tone

Out of Scope

- Real-time medical diagnosis
- Integration with hospital systems
- Clinical/legal validation

2. Proposed Solution

2.1 Key Features

- AI-powered content generation using Groq
- RAG-based retrieval using ChromaDB
- Semantic embeddings via SentenceTransformers
- Interactive UI with Streamlit
- Customizable tone, audience, and format
- Transparent display of retrieved context
- Secure API key management via .env

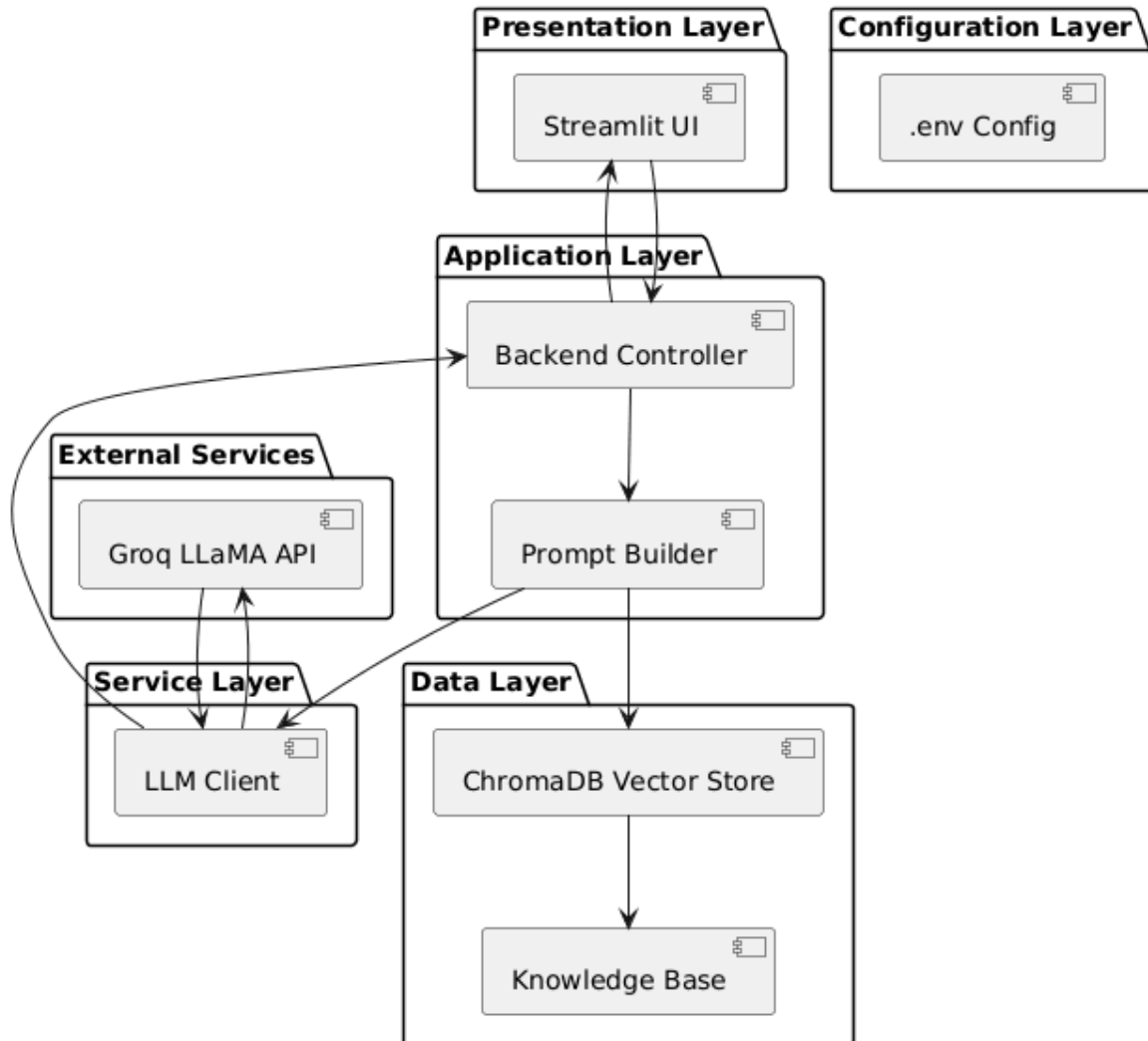
2.2 Overall Architecture

◆ Architecture Diagram

The system follows a **layered architecture**:

- **Presentation Layer:** Handles user interaction
- **Application Layer:** Controls logic and prompt construction
- **Service Layer:** Manages LLM communication
- **Data Layer:** Handles context retrieval using vector DB
- **Configuration Layer:** Stores environment variables securely

Healthcare Content Generator - System Architecture

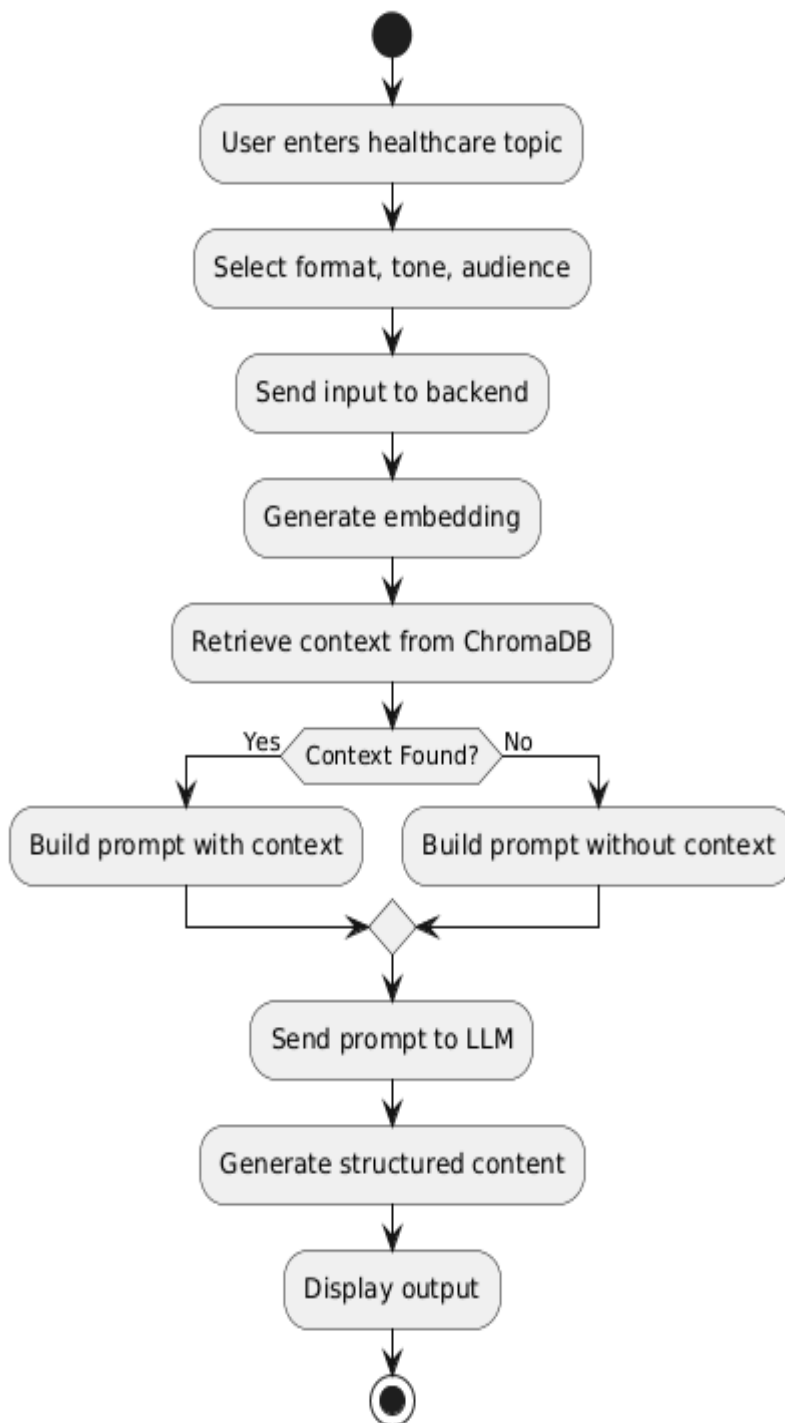


2.3 Workflow / Process Flow

Process Flow Diagram

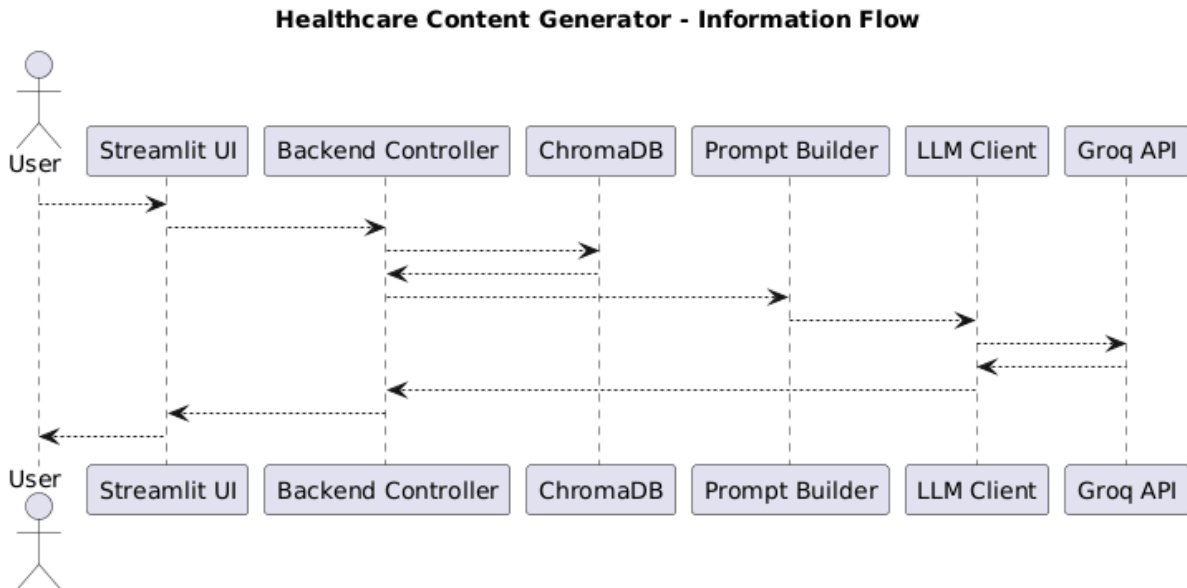
1. User provides input through UI
2. Backend processes and converts it into embeddings
3. Relevant context is retrieved from vector DB
4. Prompt is dynamically constructed
5. LLM generates structured healthcare content
6. Output is displayed with context

Healthcare Content Generator - Process Flow



2.4 Information Flow

Information Flow Diagram



2.5 Tools & Technologies Used

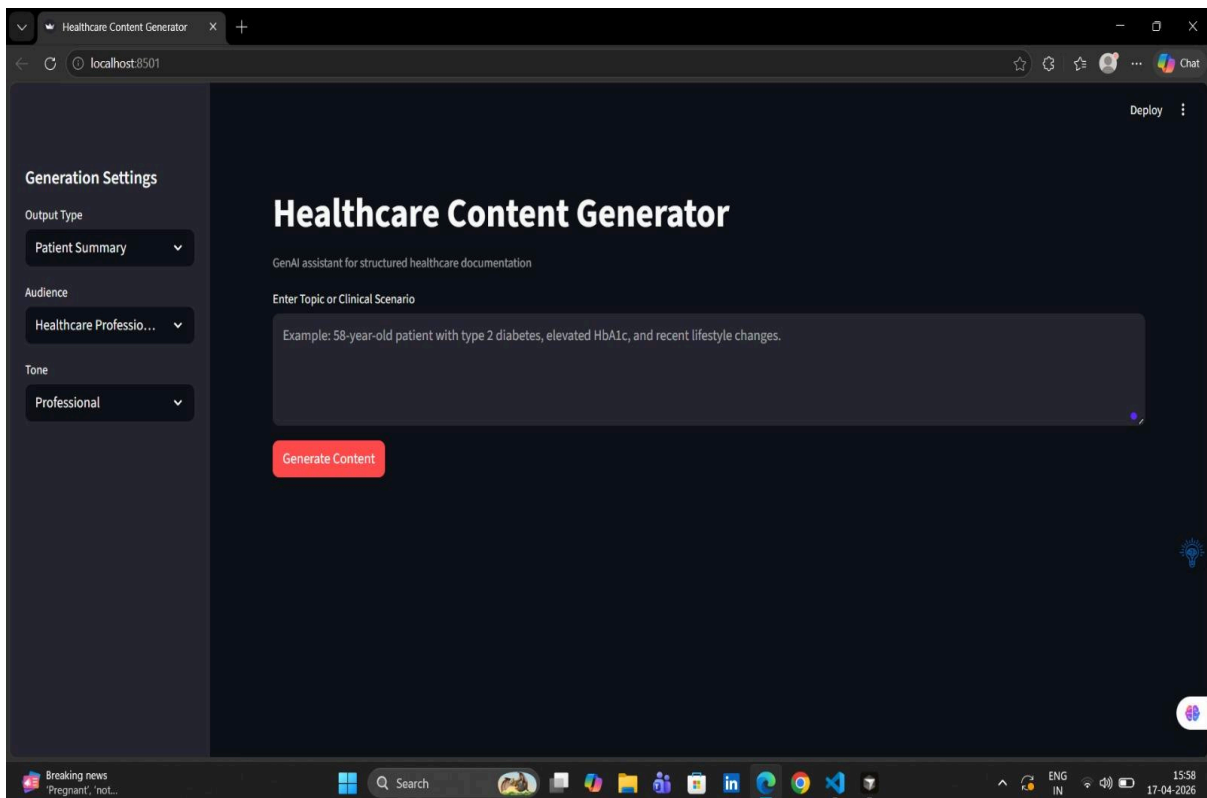
Category	Technology
Frontend	Streamlit
Backend	Python
LLM API	Groq
Vector DB	ChromaDB
Embeddings	SentenceTransformers
Version Control	GitHub

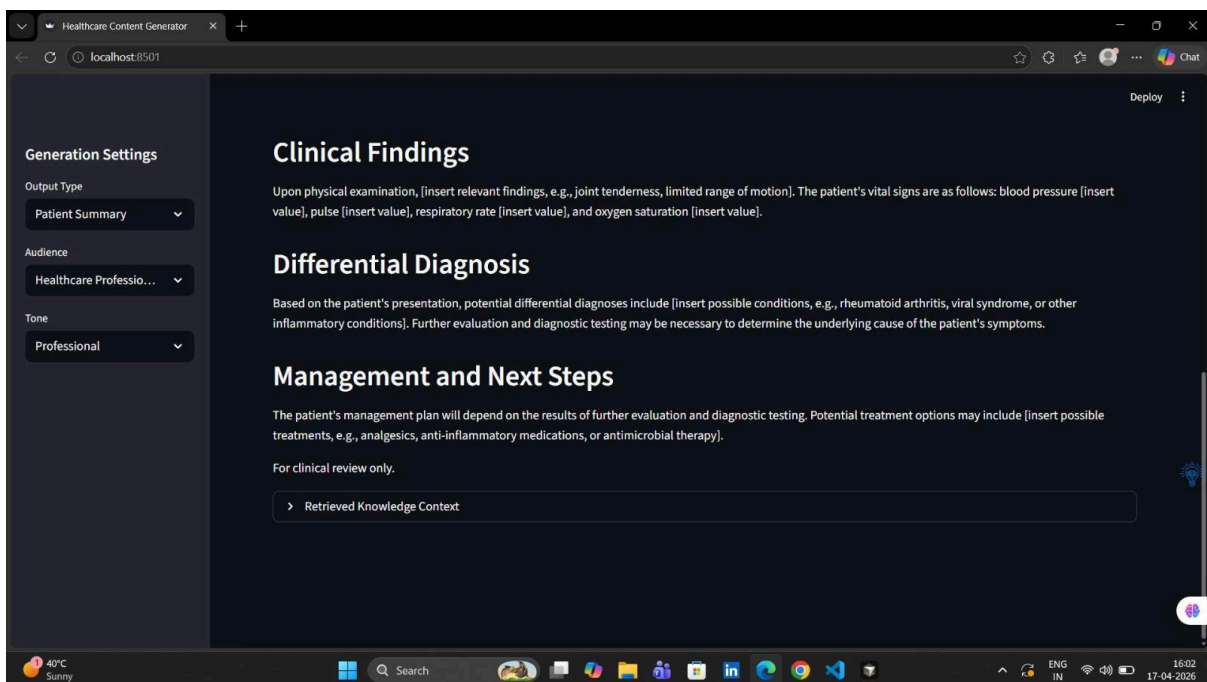
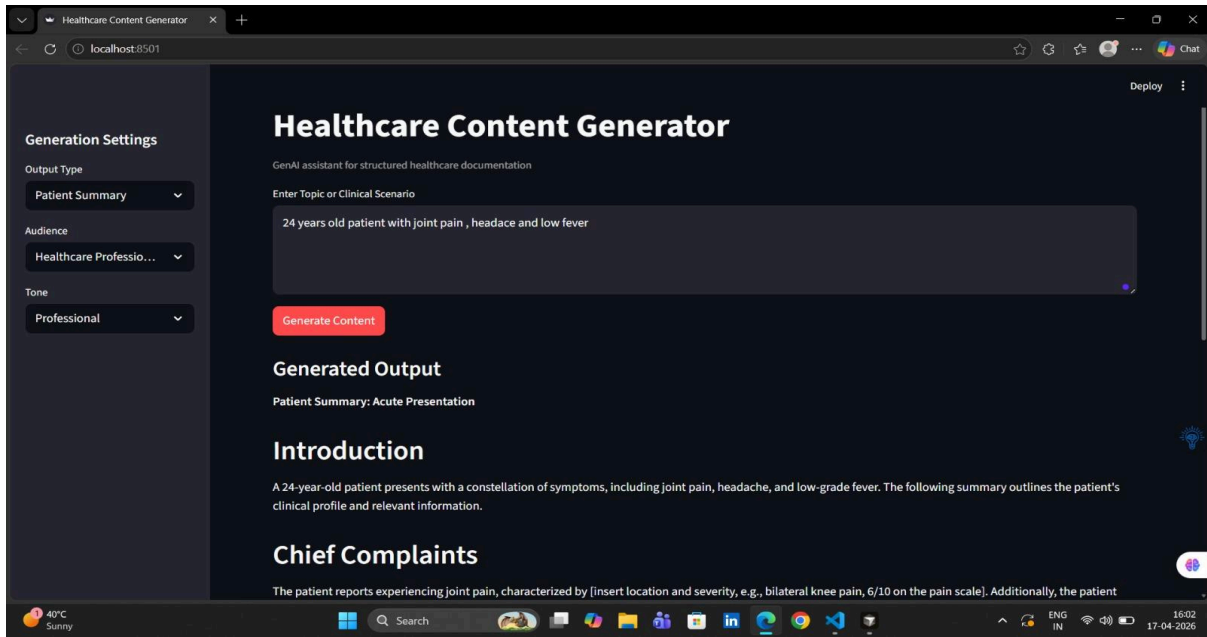
3. Results & Output

3.1 Screenshots / Outputs

The system provides:

- Input form for healthcare topic
- Output section for structured content
- Expandable context viewer





3.2 Reports / Dashboards

- Real-time generated responses
- Context transparency for validation

3.3 Key Outcomes

- Successfully built a RAG-based AI system
- Improved response accuracy using context retrieval
- Designed a modular architecture
- Reduced manual healthcare documentation effort

4. Conclusion

The project demonstrates the practical implementation of Generative AI with RAG to produce structured and context-aware healthcare content. It improves efficiency, consistency, and usability while maintaining a scalable architecture.

5. Future Scope & Enhancements

- User authentication system
- Integration with healthcare databases
- Multi-language support
- AI safety validation layer
- Deployment using Docker & cloud
- Caching with Redis
- Real-time patient history tracking

6. References

- Streamlit Documentation
- ChromaDB Documentation
- Groq API Docs
- SentenceTransformers Docs

7. GitHub

Github link - <https://github.com/Nirdesharma/Healthcare-Content-Generator>

